

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: ALPM - Prevent PM Packet

Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:
Adding a missing rule which prevents a DP IN Adapter from sending a PM Packet on the Main Link during ALPM flow if the <i>USB4 CL1 Granted through ALPM</i> bit is set to 0b.

Benefits as a result of the changes:
Prevent entering to CL1 without a grant for it from the DPTX.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
None

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Actual Change

(a). Section 10.5.7.2 Sleep Sequence

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Upon reception of an ML_PHY_SLEEP sequence, as defined in the DisplayPort Specification, a DP IN Adapter shall perform the following sleep sequence:

1. Stop sending DP Clock Sync Packets.
2. Send as Tunneled Packets, all the DP Main-Link constructs that were received prior to the ML_PHY_SLEEP.
3. Send a DP Link Control Packet of type ALPM with the following fields:
 - a. *S/W* – Set to 1b.
 - b. *SR Count* – If the DP Link is 8b/10b then set to the number of cycles as described in Section 10.5.7.1. If the DP Link is 128b/132b then set to 0h.

Note: For an 8b/10b DP Link, the DP Link Control Packet of Type ALPM may be sent by the DP IN Adapter prior to sending all the Tunneled Packets mentioned in Step 2.

4. If *USB4 CL1 Granted through ALPM* bit in the DPCD Tunnel POWER MANAGEMENT CONFIGURATION (address E0032h bit 0) is set to 1b, the DP IN Adapter shall send a PM Packet with the *CLx State* field set to 01b over the Main-Link Path. If *USB4 CL1 Granted through ALPM* bit is set to 0b, the DP IN Adapter shall not send a PM Packet with the *CLx State* field set to 01b over the Main-Link Path.

Note: A DP IN Adapter follows the sleep sequence above after receiving the first ML_PHY_SLEEP, regardless of its position in the LTTPR count.